

Coding & Solution

Date	5 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

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The **Smart Agricultural Production** project is developed to assist farmers in making informed agricultural decisions using machine learning. The application predicts the most suitable crop based on soil and environmental parameters through a web-based interface. The solution is implemented using Python and Flask, following a modular architecture that separates the user interface, business logic, machine learning model, and datasets. The application is designed to be scalable, maintainable, and user-friendly while meeting the functional requirements defined during the project planning phase.

Solution Summary

Field	Details
Repository Link / URL	Git-hub link
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Jinja2
Key Features Implemented	User-friendly web interface, Crop recommendation using Machine Learning, Soil parameter input, Prediction result display, Dataset integration, Responsive UI
Pending / Incomplete Features	Weather API integration, Fertilizer recommendation, Disease detection, User authentication, Historical prediction storage
Setup / Run Instructions	Install Python and required libraries using <code>pip install -r requirements.txt</code> , place the trained model and dataset in the project directory, then start the application using <code>python app.py</code> and access it through the local web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes